

Switching-Aware Multilayer Graph Learning for Energy-Efficient User Association and Cell Switch-Off in Dense Wireless Networks

Author Names Omitted for Draft

Abstract—Dense 5G and beyond-5G radio access networks require user association policies that reduce infrastructure energy consumption while preserving quality-of-service (QoS), load feasibility, and operational stability. Static BS switch-off can save energy, but frequent on/off transitions introduce additional cost and instability that are often ignored by snapshot-wise association policies. This paper proposes a switching-aware multilayer graph-learning framework for joint user association and BS switch-off. Each wireless snapshot is modeled as a UE-level multilayer graph with separate relation layers for association competition, interference similarity, load-pressure similarity, and temporal traffic similarity. Fixed-fusion and attention-weighted multilayer graph convolutional networks (ML-GCNs) learn UE-to-BS association preferences. A score-guided active-cell pruning stage and a stability-aware pruning stage then convert learned preferences into compact active-BS sets under QoS/load repair. The framework is analyzed in terms of permutation equivariance, aggregation-induced information loss, feasibility-preserving repair, and switching-adjusted energy. Ten-seed simulation results show that greedy sleep control is the strongest non-learning policy for static snapshot energy, achieving 59.7% mean energy saving relative to an all-on network. However, when BS switching cost is included, the proposed stable ML-GCN improves switching-adjusted energy saving to 50.4%, compared with 48.4% for greedy sleep, while maintaining a served ratio of 1.0 and reducing BS state changes by approximately 0.95 per snapshot. Runtime measurements further show that stable learned policies have lower online latency than greedy sleep and much lower latency than finite active-cell search. These results support multilayer graph learning combined with stability-aware pruning as a practical framework for feasible, energy-efficient, and switching-aware wireless control.

Index Terms—Graph neural networks, multilayer networks, user association, network energy savings, switching cost, 6G, base-station switch-off.

I. INTRODUCTION

DENSE radio access networks are expected to support rapidly increasing traffic demand while reducing the energy footprint of wireless infrastructure. This tension is especially visible in dense 5G and beyond-5G deployments, where operators deploy macro, micro, pico, and indoor small-cell layers to improve coverage and hotspot capacity. Such densification improves service during busy periods, but it also creates many lightly loaded or temporarily redundant cells when traffic is spatially uneven or time varying. Prior green HetNet studies therefore treat base-station (BS) ON/OFF control as a radio-resource-management problem: redundant small cells may be switched to a low-power state, and their users may be served by nearby active BSs, provided rate and

load constraints remain satisfied [1]–[3]. This paper follows that infrastructure-energy viewpoint. The BSs are not assumed to be battery-limited devices; instead, the radio access network (RAN) is an energy-cost- and carbon-sensitive infrastructure whose active radio units, power amplifiers, cooling systems, and baseband resources should be used only when they provide service value [4], [5].

The relevant operational question is therefore not simply which BS gives each UE the strongest signal. It is a joint control question: which BSs should remain active, which UEs should be associated with them, and how should the active-cell set evolve over time without violating service constraints. This decision is naturally made by a network-side control entity, such as a centralized SON function, cloud/edge RAN controller, or O-RAN RIC application, using UE measurements, channel estimates, traffic demand, BS load, and previous BS states. O-RAN energy-saving work explicitly identifies RF-channel reconfiguration, advanced sleep modes, and cell/carrier shutdown as network energy-saving features coordinated through RIC/management interfaces [5]. In this setting, a BS sleep command is meaningful only if users that would otherwise be served by that BS can be reassociated to active cells while maintaining radio feasibility, rate demand, and load limits.

This coupling makes energy-aware user association more difficult than association-only control. Since a BS consumes non-negligible static power even when lightly loaded, energy-efficient network operation requires the controller to reason about the global consequence of activating or deactivating cells. It also requires temporal stability. A policy that independently minimizes energy at each snapshot can increase BS state changes, and these transitions may incur signaling, hardware, and service-continuity costs. Energy-efficient control should therefore be evaluated not only by static power reduction, but also by the stability of the active-cell sequence.

User association has therefore become a central control problem in dense wireless networks. Classical association rules based on received signal strength, signal-to-interference-plus-noise ratio (SINR), or load balancing are computationally attractive, but they are fundamentally local. They typically assign each UE using the quality or load of a small number of candidate BSs and do not explicitly reason about the global energy consequence of activating or deactivating BSs. Optimization-based formulations can model the joint association and switch-off problem more directly, but the resulting mixed discrete-continuous decision space is difficult

to solve repeatedly at network time scales. Learning-based methods are a natural alternative because they can amortize computation after training; however, the quality of a learned policy depends strongly on the graph representation through which the wireless system is presented to the model.

Graph neural networks (GNNs) are attractive for wireless resource management because wireless networks are relational systems. A UE is not an isolated sample: its best association depends on neighboring UEs, competing BSs, interference coupling, traffic demand, and load created by other association decisions. Recent wireless GNN studies show that graph-based models can provide scalable approximations for resource-allocation tasks and can exploit permutation structure in wireless graphs [6], [7]. Nevertheless, many graph-learning formulations reduce the system to a single relation graph. Such a collapse is convenient, but it can hide an important structural distinction: a dense wireless snapshot contains multiple kinds of relations whose semantics are not interchangeable. Two UEs may compete for the same serving BS, exhibit similar traffic demand, experience correlated interference conditions, or follow similar temporal traffic patterns. These relations may overlap, but they are not equivalent and should not necessarily be represented by a single averaged graph.

This observation motivates a multilayer network formulation. In multilayer network models, a system is represented not only by nodes and edges, but also by layers that encode different interaction types, time slices, modalities, or couplings [8]–[10]. This distinction is not only descriptive. Prior work on multilayer networks has shown that aggregation can remove structural information, distort centrality rankings, and obscure layer-specific organization [11], [12]. For energy-aware wireless control, this issue is consequential. If association competition, interference, load pressure, and temporal traffic similarity are collapsed into one graph, a learning model may receive an averaged relation that no longer identifies which physical or operational mechanism caused a dependency between UEs. A policy trained on such an aggregate graph may therefore associate users correctly in easy cases but fail to learn why certain BSs can be switched off without violating quality-of-service (QoS) and load constraints.

In this paper, we study energy-efficient user association and BS switch-off as a multilayer graph decision problem. For each wireless snapshot, we construct a UE-level multilayer graph

$$\mathcal{G} = \left(\mathcal{V}, \mathbf{X}, \{\mathbf{A}^{(\ell)}\}_{\ell \in \mathcal{L}} \right), \quad (1)$$

where $\mathcal{V}$ is the set of UEs, $\mathbf{X}$ contains UE radio and traffic features, and each layer $\mathbf{A}^{(\ell)}$ represents a distinct wireless relation. The layer set $\mathcal{L}$ is defined to preserve four operational dependencies that are relevant to energy-aware association:

- the *association layer*, which connects UEs that compete for the same strongest or candidate BS;
- the *interference layer*, which links UEs with similar channel profiles and interference exposure;
- the *load layer*, which captures demand similarity and potential load-pressure coupling among UEs; and
- the *temporal layer*, which represents spatial-traffic similarity under time-varying demand.

Fig. 2 illustrates how these relation layers are constructed, processed by layer-specific graph filters, and fused before association and active-cell decisions. The final decision is evaluated by network energy, QoS satisfaction, BS activity, and load feasibility rather than by assignment accuracy alone.

The central research question is:

Can preserving multilayer wireless structure improve energy-aware user association and BS switch-off decisions, compared with conventional association heuristics, sleep-control baselines, and single-layer graph-learning models, while maintaining QoS, load feasibility, and active-cell stability?

The proposed framework is intended for execution at the network controller, not at individual UEs. The controller observes each traffic snapshot, constructs the multilayer UE graph, predicts association preferences, selects a compact active-BS set, and then applies a feasibility repair before the decision is sent to the RAN as association and sleep-mode control. The framework has two linked components. First, a multilayer GNN predicts UE-to-BS association preferences using layer-specific graph propagation. Second, an active-cell selector uses the learned association distribution and BS-level radio/load summaries to identify a compact active BS set. A QoS- and load-aware repair procedure then converts the learned preferences into a feasible association whenever possible, activating additional BSs only when the predicted active set cannot support the traffic realization. This design separates representation learning from constraint enforcement: the GNN learns multilayer structure, while the repair stage protects wireless feasibility. We further introduce score-guided pruning and stability-aware pruning policies. The score-guided policy removes low-score BSs only when energy and feasibility improve. The stability-aware policy uses the previous snapshot's active set as inertia and accepts a BS state change only when the switching-adjusted objective improves or feasibility requires the transition.

The paper makes the following contributions.

- 1) We formulate joint user association and BS switch-off as a multilayer network decision problem, with separate layers for association competition, interference coupling, load pressure, and temporal traffic similarity. The formulation preserves layer-specific wireless dependencies and admits theoretical analysis of permutation equivariance, aggregation-induced information loss, and feasibility-preserving decision repair.
- 2) We develop fixed-fusion and attention-weighted multilayer GNN models for UE association preferences and combine them with score-guided and stability-aware active-cell pruning. The resulting policies explicitly trade off static BS energy, switching cost, QoS satisfaction, and load feasibility.
- 3) We define a comparative metric suite for constrained wireless control, including static energy saving, switching-adjusted energy saving, energy gap relative to a finite-search reference, served ratio, active BS count, BS state changes, maximum load, and mean active load. Layer ablation, attention entropy, relation-redundancy,

and runtime measurements are included to assess modeling and operational tradeoffs.

- 4) We conduct a simulation-based comparative study against conventional RSRP, SINR, and load-aware association heuristics, explicit sleep-control heuristics, greedy BS sleep control, finite active-cell search, feature-only learning, aggregated GCN learning, fixed multilayer GCN learning, and attention-weighted multilayer GCN learning. The results distinguish static-energy performance from switching-aware operation.

The remainder of this paper is organized as follows. Section II reviews wireless user association, graph learning for resource management, and multilayer graph modeling. Section III presents the system model and energy-aware association objective. Section IV develops the proposed multilayer graph-learning framework and its mathematical properties. Section V describes the simulation design, baselines, and metrics. Section VI reports and discusses the comparative results. Section VII concludes the paper.

II. RELATED WORK

Energy-aware user association has been widely studied as a mechanism for improving spectral efficiency, load distribution, and infrastructure energy consumption in dense wireless networks. Conventional association rules based on received signal strength, SINR, or load provide simple and scalable decisions, but they are usually myopic with respect to BS switch-off because the activation cost of each BS is not directly optimized. Optimization-based methods can incorporate energy and service constraints explicitly, but the joint association and activation problem is combinatorial and must be solved under changing channel and traffic conditions.

BS ON/OFF switching has a long history in green cellular networking. Dynamic switching methods exploit the fact that cellular traffic varies across time and space, so BSs dimensioned for peak demand may be underutilized during off-peak periods [2]. In HetNets, redundant small cells can be placed in sleep mode and users can be offloaded to active macro, small-cell, or femtocell resources when QoS remains feasible [1]. Surveys on BS ON/OFF switching in 5G further emphasize that dense, heterogeneous deployments create energy-saving opportunities, but also introduce coverage, interference, handover, and coordination challenges [3]. The present paper uses this literature as the physical motivation for the active-cell decision: BS switch-off is not a UE-side battery-saving action, but an infrastructure energy-management action applied by the RAN to reduce static and load-dependent network power.

BS switch-off methods are often evaluated snapshot by snapshot, but practical cell activation also has a temporal dimension. Repeatedly changing the active BS set can introduce signaling overhead, hardware transition cost, and service instability. The present work therefore treats BS state changes as an explicit metric and evaluates switching-adjusted energy in addition to static network power. This distinction is central to the comparison with greedy sleep-control baselines.

GNNs provide a learning-based alternative for resource-management problems in which the wireless topology varies

across network realizations. Random-edge GNNs have been used to approximate wireless resource allocation over changing interference graphs [6], while scalable GNN architectures have been analyzed for radio resource management with permutation properties and size generalization [7]. Graph attention has also been applied to wireless user association [13]. These studies support the use of graph learning for wireless control, but they typically represent the network through a single graph or a single dominant relation.

Multilayer network models provide a formal way to represent systems with multiple edge semantics, time slices, or interaction channels [8]–[10]. The importance of preserving layer structure is supported by work on structural reducibility and multilayer node ranking, where aggregation can remove relevant information or change structural rankings [11], [12]. These results are relevant to dense wireless networks because association competition, interference, load, and temporal traffic similarity describe different dependencies among UEs.

Several recent GNN studies have extended graph learning to multilayer or multi-view settings. The mGNN framework generalizes GNN operations to multilayer networks [14]. Dynamic-multilayer GCNs have been used to combine intralayer and interlayer convolution for industrial monitoring [15], while attention-based dynamic multilayer GNNs have been developed for temporal multilayer prediction tasks [16]. MVMA-GCN further uses multi-view attention and relationship dissimilarity to learn from multiple node-relation graphs [17]. These works demonstrate the value of layer-aware graph learning, but they do not address energy-aware wireless association with active-cell control. The present work focuses on this setting by coupling a multilayer UE representation with BS switch-off, QoS/load repair, and energy-oriented evaluation.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network Snapshot

Consider a dense downlink control region managed by a network-side controller. The region contains a set of controllable serving cells $\mathcal{B} = \{1, \dots, B\}$ and a set of UEs $\mathcal{U} = \{1, \dots, N\}$ observed at traffic snapshot t . In a HetNet, the controllable set may correspond to small cells under an overlay macro layer; in a dense single-tier deployment, it may correspond to a cluster of neighboring BSs. In both cases, the abstraction is the same: the controller can choose an active subset of BSs and can reassociate UEs among active candidates, subject to radio, rate, and load feasibility. The position, traffic demand, and radio measurements of UE i are represented by a feature vector $\mathbf{x}_i \in \mathbb{R}^F$, and the snapshot feature matrix is $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times F}$. Let $d_i(t)$ denote the traffic demand of UE i in Mbps. The large-scale and small-scale channel gain from BS b to UE i is denoted by $g_{i,b}(t)$, and the BS transmit power is P_{tx} .

The user association and BS switch-off decision is described by two binary variables. Let

$$z_{i,b} = \begin{cases} 1, & \text{if UE } i \text{ is associated with BS } b, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

and let $a_b \in \{0, 1\}$ indicate whether BS b is active. Each UE is associated with exactly one BS:

$$\sum_{b \in \mathcal{B}} z_{i,b} = 1, \quad \forall i \in \mathcal{U}, \quad (3)$$

and a UE can be assigned only to an active BS:

$$z_{i,b} \leq a_b, \quad \forall i \in \mathcal{U}, b \in \mathcal{B}. \quad (4)$$

B. SINR, Rate, and Load

For a given active-BS vector $\mathbf{a} = [a_1, \dots, a_B]^T$, the downlink SINR that UE i would obtain from BS b is

$$\gamma_{i,b}(\mathbf{a}) = \frac{a_b P_{\text{tx}} g_{i,b}}{\sigma^2 + \sum_{c \in \mathcal{B}, c \neq b} a_c P_{\text{tx}} g_{i,c}}, \quad (5)$$

where σ^2 is the receiver noise power. The corresponding achievable rate is modeled as

$$R_{i,b}(\mathbf{a}) = W \log_2 (1 + \gamma_{i,b}(\mathbf{a})), \quad (6)$$

where W is the system bandwidth. The RSRP feasibility condition is

$$P_{\text{tx}} g_{i,b} \geq \Gamma_{\text{rsrp}}, \quad (7)$$

and the minimum-rate condition is

$$R_{i,b}(\mathbf{a}) \geq R_{\min}. \quad (8)$$

The load contribution of UE i to BS b is the fraction of radio resources needed to support its demand:

$$\ell_{i,b}(\mathbf{a}) = \frac{d_i}{R_{i,b}(\mathbf{a})}. \quad (9)$$

The total load of BS b under association matrix $\mathbf{Z}$ is

$$\rho_b(\mathbf{Z}, \mathbf{a}) = \sum_{i \in \mathcal{U}} z_{i,b} \ell_{i,b}(\mathbf{a}), \quad (10)$$

and the load constraint is

$$\rho_b(\mathbf{Z}, \mathbf{a}) \leq \rho_{\max}, \quad \forall b \in \mathcal{B}. \quad (11)$$

C. Energy Model

The network energy model separates the static cost of keeping a BS active, the load-dependent cost of serving traffic, and the sleep-mode power of inactive BSs. This follows the BS power-modeling view used in green cellular-network studies, where sleep mode reduces but does not necessarily eliminate site power because some circuitry, cooling, monitoring, or wake-up functionality may remain active [1], [2]. The power consumed by BS b is

$$P_b(\mathbf{Z}, \mathbf{a}) = a_b (P_{\text{fix}} + P_{\text{dyn}} \rho_b) + (1 - a_b) P_{\text{sleep}}, \quad (12)$$

where P_{fix} is the fixed active power, P_{dyn} is the dynamic load-dependent power coefficient, and P_{sleep} is the sleep-mode power. The total network power is

$$P_{\text{net}}(\mathbf{Z}, \mathbf{a}) = \sum_{b \in \mathcal{B}} P_b(\mathbf{Z}, \mathbf{a}). \quad (13)$$

The all-on reference power is obtained by setting $a_b = 1$ for all $b \in \mathcal{B}$. This reference is used to report energy saving:

$$\text{ES} = \frac{P_{\text{all}} - P_{\text{net}}}{P_{\text{all}}}. \quad (14)$$

For temporally ordered snapshots, static energy alone is incomplete because active-cell transitions can introduce operational cost. Let $\mathbf{a}(t)$ be the active-BS vector at snapshot t . The number of BS state changes is

$$S(t) = \|\mathbf{a}(t) - \mathbf{a}(t-1)\|_0, \quad (15)$$

where $S(1) = 0$ for the first evaluated snapshot. With a per-transition switching cost κ , the switching-adjusted power is

$$P_{\text{sw}}(t) = P_{\text{net}}(t) + \kappa S(t). \quad (16)$$

The corresponding switching-adjusted energy saving is

$$\text{ES}_{\text{sw}} = \frac{P_{\text{all}} - P_{\text{sw}}}{P_{\text{all}}}. \quad (17)$$

When $\kappa = 0$, (17) reduces to the static energy-saving metric in (14).

D. Optimization Problem

The ideal energy-aware association problem is to minimize network power while satisfying association, activation, QoS, and load constraints:

$$\begin{aligned} \min_{\mathbf{Z}, \mathbf{a}} \quad & P_{\text{net}}(\mathbf{Z}, \mathbf{a}) \\ \text{s.t.} \quad & \sum_{b \in \mathcal{B}} z_{i,b} = 1, & \forall i \in \mathcal{U}, \\ & z_{i,b} \leq a_b, & \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & z_{i,b} P_{\text{tx}} g_{i,b} \geq z_{i,b} \Gamma_{\text{rsrp}}, & \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & z_{i,b} R_{i,b}(\mathbf{a}) \geq z_{i,b} R_{\min}, & \forall i \in \mathcal{U}, b \in \mathcal{B}, \\ & \rho_b(\mathbf{Z}, \mathbf{a}) \leq \rho_{\max}, & \forall b \in \mathcal{B}, \\ & z_{i,b} \in \{0, 1\}, a_b \in \{0, 1\}. \end{aligned} \quad (18)$$

This problem is combinatorial because the association matrix and active-BS vector are discrete and coupled through SINR, rate, load, and power. The goal of the proposed learning framework is therefore not to solve (18) exactly at every snapshot, but to learn a scalable policy that approximates low-energy feasible decisions from the multilayer wireless graph representation. In the switching-aware setting, the same feasible association variables are evaluated using $P_{\text{sw}}(t)$ in (16), which makes the active-cell sequence $\{\mathbf{a}(t)\}$ part of the operational objective.

IV. PROPOSED MULTILAYER GRAPH LEARNING FRAMEWORK

A. Overview

The proposed framework converts each wireless snapshot into a multilayer UE graph, learns UE-to-BS association preferences using layer-specific graph filters, predicts a compact active-BS set, and applies a QoS/load repair step before evaluation. Fig. 1 gives the computational pipeline, while Fig. 2 gives the detailed multilayer model architecture. The key modeling choice is that different wireless relations are not merged before graph propagation. Instead, each relation layer is processed by a dedicated graph filter and fused only after layer-wise representation learning.

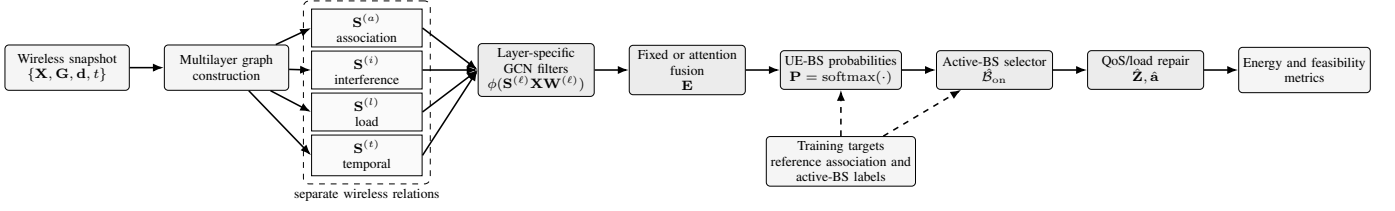


Fig. 1. Computational pipeline of the proposed framework. Wireless snapshots are converted into separate relation layers, processed by layer-specific GCN filters, fused into UE embeddings, and passed to association, active-BS selection, repair, and evaluation modules.

Proposed Multilayer Graph Learning Framework for Energy-Aware User Association

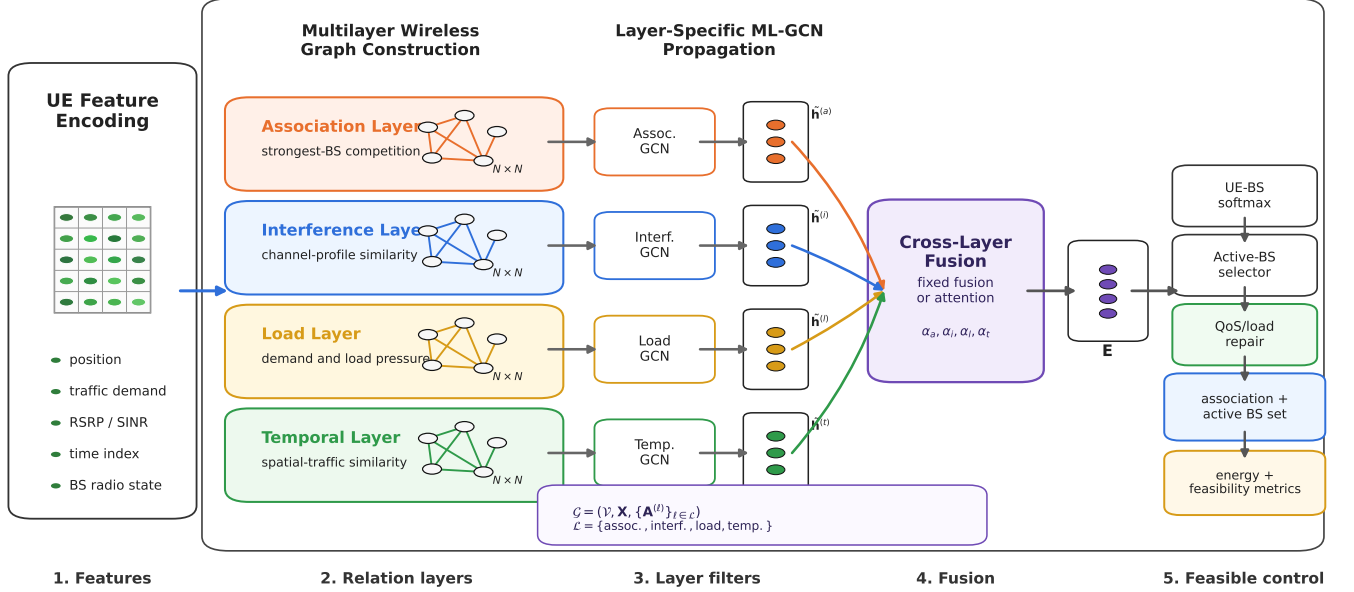


Fig. 2. Proposed multilayer graph-learning framework for energy-aware user association and BS switch-off. Separate wireless relation layers are preserved during graph propagation and fused before association, active-BS selection, and QoS/load repair.

B. Multilayer Wireless Graph Construction

For each snapshot, let $\mathbf{G} \in \mathbb{R}^{N \times B}$ denote the channel-gain matrix with entries $g_{i,b}$, and let $\mathbf{r}_i$ be the position of UE i . The proposed model builds a multilayer graph

$$\mathcal{M} = (\mathcal{U}, \mathbf{X}, \{\mathbf{S}^{(\ell)}\}_{\ell \in \mathcal{L}}), \quad \mathcal{L} = \{a, i, l, t\}, \quad (19)$$

where a, i, l , and t denote association, interference, load, and temporal layers, respectively. Each layer is constructed as a nonnegative adjacency matrix $\mathbf{A}^{(\ell)}$ and converted into a row-stochastic support matrix

$$\mathbf{S}^{(\ell)} = (\mathbf{D}^{(\ell)})^{-1} (\mathbf{A}^{(\ell)} + \mathbf{I}), \quad (20)$$

where $\mathbf{D}^{(\ell)}$ is the diagonal row-sum matrix of $\mathbf{A}^{(\ell)} + \mathbf{I}$. The identity term preserves each UE's own features during propagation.

The association layer connects UEs that compete for the same strongest BS. Let

$$b_i^* = \arg \max_{b \in \mathcal{B}} g_{i,b}. \quad (21)$$

Then

$$A_{j,k}^{(a)} = \mathbf{1}\{b_j^* = b_k^*\}, \quad j \neq k. \quad (22)$$

The interference layer compares normalized channel-gain profiles. Define $\mathbf{q}_j = \mathbf{g}_j / \|\mathbf{g}_j\|_1$, where $\mathbf{g}_j$ is the j th row of $\mathbf{G}$. The interference similarity is

$$A_{j,k}^{(i)} = \mathbf{q}_j^T \mathbf{q}_k, \quad j \neq k. \quad (23)$$

The load layer links UEs with similar traffic demand within the same association region:

$$A_{j,k}^{(l)} = A_{j,k}^{(a)} \exp\left(-\eta_l \frac{|d_j - d_k|}{d_{\max}}\right), \quad (24)$$

where $\eta_l > 0$ controls demand-similarity sharpness and $d_{\max}$ is the maximum demand scale. Finally, the temporal layer captures spatial-demand similarity modulated by the traffic period:

$$A_{j,k}^{(t)} = h(t) \exp\left(-\eta_r \frac{\|\mathbf{r}_j - \mathbf{r}_k\|_2}{R}\right) \exp\left(-\eta_d \frac{|d_j - d_k|}{d_{\max}}\right), \quad (25)$$

where R is the cell-area radius and $h(t)$ is a bounded periodic traffic factor. Sparse top- K thresholding is applied to the similarity layers to avoid dense message passing and to emphasize dominant local dependencies.

C. Layer-Aware Association Learning

Given $\mathbf{X}$ and the supports $\{\mathbf{S}^{(\ell)}\}$, the fixed-fusion multi-layer GCN computes one hidden representation per layer:

$$\mathbf{H}^{(\ell)} = \phi\left(\mathbf{S}^{(\ell)}\mathbf{X}\mathbf{W}^{(\ell)}\right), \quad \ell \in \mathcal{L}, \quad (26)$$

where $\phi(\cdot)$ is a pointwise nonlinearity and $\mathbf{W}^{(\ell)}$ is a layer-specific trainable matrix. The layer-wise embeddings are fused as

$$\mathbf{E} = \left[\mathbf{H}^{(a)} \parallel \mathbf{H}^{(i)} \parallel \mathbf{H}^{(l)} \parallel \mathbf{H}^{(t)}\right], \quad (27)$$

where $\parallel$ denotes feature concatenation. The UE-to-BS association probability matrix is

$$\mathbf{P} = \text{softmax}(\mathbf{E}\mathbf{W}_o + \mathbf{b}_o), \quad (28)$$

where $P_{i,b}$ is the learned probability that UE i should be associated with BS b .

The attention-weighted model assigns a trainable nonnegative importance weight to each layer. Let η_ℓ be the attention logit of layer ℓ . The normalized layer weight is

$$\alpha_\ell = \frac{\exp(\eta_\ell)}{\sum_{r \in \mathcal{L}} \exp(\eta_r)}. \quad (29)$$

The fused embedding becomes

$$\mathbf{E}_{\text{att}} = \left[|\mathcal{L}|\alpha_a\mathbf{H}^{(a)} \parallel |\mathcal{L}|\alpha_i\mathbf{H}^{(i)} \parallel |\mathcal{L}|\alpha_l\mathbf{H}^{(l)} \parallel |\mathcal{L}|\alpha_t\mathbf{H}^{(t)}\right]. \quad (30)$$

The factor $|\mathcal{L}|$ keeps the scale comparable to fixed fusion when the attention distribution is uniform.

The association predictor is trained by supervised imitation of low-energy reference decisions:

$$\mathcal{L}_{\text{ua}} = -\frac{1}{N} \sum_{i \in \mathcal{U}} \sum_{b \in \mathcal{B}} y_{i,b} \log P_{i,b}, \quad (31)$$

where $y_{i,b}$ is the reference association label. This loss trains the GNN to learn association preferences; feasibility and switch-off behavior are handled by the active-cell and repair stages below.

D. Active-BS Selection and QoS/Load Repair

Association probabilities alone do not determine which BSs should remain active, because a model may assign every UE while still keeping unnecessary BSs on. Therefore, the proposed framework uses a second-stage active-BS selector. For each BS b , a summary vector ψ_b is formed from probability statistics, demand-weighted probability mass, approximate load, channel quality, and distance statistics. The active score is

$$s_b = \sigma\left(\theta^T \bar{\psi}_b + c\right), \quad (32)$$

where $\bar{\psi}_b$ is the standardized BS summary vector and $\sigma(\cdot)$ is the logistic sigmoid. The selected active set is the set of top-scoring BSs,

$$\hat{\mathcal{B}}_{\text{on}} = \text{TopK}_{b \in \mathcal{B}}(s_b). \quad (33)$$

The value of K is estimated from the average active-BS count of the reference solutions in the training set.

The fixed top- K selector is useful as a baseline, but it can be too rigid because different snapshots require different active-cell counts. We therefore define two pruning policies. The score-guided pruning policy starts from the all-on set and evaluates candidate removals in increasing order of s_b . A BS b is deactivated only if the repaired association remains feasible and the static energy in (13) does not increase. The accepted set is

$$\hat{\mathcal{A}}_{\text{pr}} = \text{Prune}(\mathcal{B}, \mathbf{s}, P_{\text{net}}, \Phi), \quad (34)$$

where Φ is the feasibility predicate. This policy adapts the active count per snapshot but does not use temporal inertia.

The stability-aware policy extends pruning to ordered snapshots. Given the previous active set $\hat{\mathcal{A}}(t-1)$, it first repairs the current snapshot using this set when feasible, then compares candidate deactivations and the independently pruned set under the switching-adjusted objective

$$J_{\text{sw}}(\mathcal{A}, t) = P_{\text{net}}(\mathcal{A}, t) + \kappa \|\mathbf{a}_{\mathcal{A}} - \mathbf{a}(t-1)\|_0. \quad (35)$$

A state transition is accepted only if it improves (35) while preserving QoS and load feasibility, or if activating additional BSs is required to satisfy feasibility. This mechanism directly penalizes unnecessary active-cell churn.

For any candidate active set, the initial UE assignment chooses the most likely BS among the active candidates. A repair routine then checks the RSRP, rate, and load constraints in (7)–(11). If the selected active set cannot support the snapshot, additional BSs are activated according to their aggregate gain-demand score over stressed UEs. Algorithm 1 gives the complete fixed top- K inference procedure; score-guided and stability-aware pruning use the same repair operator with different active-set proposals.

E. Mathematical Properties

Let $\mathfrak{M} = (\mathbf{X}, \{\mathbf{S}^{(\ell)}\}_{\ell \in \mathcal{L}})$ denote the multilayer input and define the layer-aware encoder F_Θ by (26)–(30). For a permutation matrix Π , define the action

$$T_\Pi(\mathfrak{M}) = \left(\Pi\mathbf{X}, \{\Pi\mathbf{S}^{(\ell)}\Pi^T\}_{\ell \in \mathcal{L}}\right). \quad (36)$$

Define also the feasibility predicate

$$\Phi(\mathbf{Z}, \mathbf{a}) = 1 \iff (3), (4), (7), (8), (11) \text{ hold for } (\mathbf{Z}, \mathbf{a}). \quad (37)$$

Theorem 1 (Equivariance of the multilayer encoder). *For any permutation matrix Π ,*

$$F_\Theta(T_\Pi(\mathfrak{M})) = \Pi F_\Theta(\mathfrak{M}). \quad (38)$$

Proof. Let $\mathbf{H}^{(\ell)}$ and $\tilde{\mathbf{H}}^{(\ell)}$ be the layer- ℓ hidden matrices generated from $\mathfrak{M}$ and $T_\Pi(\mathfrak{M})$, respectively. Then

$$\begin{aligned} \tilde{\mathbf{H}}^{(\ell)} &= \phi\left(\Pi\mathbf{S}^{(\ell)}\Pi^T\Pi\mathbf{X}\mathbf{W}^{(\ell)}\right) \\ &= \phi\left(\Pi\mathbf{S}^{(\ell)}\mathbf{X}\mathbf{W}^{(\ell)}\right) = \Pi\phi\left(\mathbf{S}^{(\ell)}\mathbf{X}\mathbf{W}^{(\ell)}\right) \\ &= \Pi\mathbf{H}^{(\ell)}. \end{aligned} \quad (39)$$

Algorithm 1 Multilayer GNN Association With Active-BS Repair

Require: $(\mathbf{X}, \mathbf{G}, \mathbf{d}, t)$, parameters $\Theta = \{\mathbf{W}^{(\ell)}, \mathbf{W}_o, \mathbf{b}_o, \eta_\ell\}$, selector (θ, c) , K

Ensure: $\hat{\mathbf{Z}}, \hat{\mathbf{a}}$

```

1:  $\mathcal{L} \leftarrow \{a, i, l, t\}$ 
2:  $\mathbf{S}^{(\ell)} \leftarrow \text{LayerSupport}_\ell(\mathbf{X}, \mathbf{G}, \mathbf{d}, t)$ ,  $\ell \in \mathcal{L}$ 
3: for  $\ell \in \mathcal{L}$  do
4:    $\mathbf{H}^{(\ell)} \leftarrow \phi(\mathbf{S}^{(\ell)} \mathbf{X} \mathbf{W}^{(\ell)})$ 
5: end for
6:  $\alpha_\ell \leftarrow \exp(\eta_\ell) / \sum_{r \in \mathcal{L}} \exp(\eta_r)$ 
7:  $\mathbf{E} \leftarrow \|\ell \in \mathcal{L}\| (|\mathcal{L}| \alpha_\ell \mathbf{H}^{(\ell)})$ 
8:  $\mathbf{P} \leftarrow \text{softmax}(\mathbf{E} \mathbf{W}_o + \mathbf{b}_o)$ 
9: for  $b \in \mathcal{B}$  do
10:   $\psi_b \leftarrow \text{BSFeatures}(b, \mathbf{P}, \mathbf{G}, \mathbf{d})$ 
11:   $s_b \leftarrow \sigma(\theta^T \psi_b + c)$ 
12: end for
13:  $\mathcal{A} \leftarrow \text{TopK}_{b \in \mathcal{B}}(s_b, K)$ 
14: repeat
15:   $z_{i,b} \leftarrow 0, \forall i, b$ 
16:  for  $i \in \mathcal{U}$  do
17:     $\mathcal{C}_i \leftarrow \{b \in \mathcal{A} : R_{i,b}(\mathcal{A}) \geq R_{\min}, P_{\text{tx}} g_{i,b} \geq \Gamma_{\text{rsrp}}\}$ 
18:    if  $\mathcal{C}_i = \emptyset$  then
19:       $\mathcal{C}_i \leftarrow \mathcal{A}$ 
20:    end if
21:     $b_i \leftarrow \arg \max_{b \in \mathcal{C}_i} P_{i,b}$ 
22:     $z_{i,b_i} \leftarrow 1$ 
23:  end for
24:   $\rho_b \leftarrow \sum_{i \in \mathcal{U}} z_{i,b} d_i / R_{i,b}(\mathcal{A})$ 
25:   $\mathcal{V} \leftarrow \{i : R_{i,b_i} < R_{\min} \text{ or } P_{\text{tx}} g_{i,b_i} < \Gamma_{\text{rsrp}} \text{ or } \rho_{b_i} > \rho_{\max}\}$ 
26:  if  $\mathcal{V} = \emptyset$  or  $|\mathcal{A}| = B$  then
27:    break
28:  end if
29:   $b^+ \leftarrow \arg \max_{b \in \mathcal{B} \setminus \mathcal{A}} \sum_{i \in \mathcal{V}} d_i g_{i,b}$ 
30:   $\mathcal{A} \leftarrow \mathcal{A} \cup \{b^+\}$ 
31: until  $\mathcal{V} = \emptyset$  or  $|\mathcal{A}| = B$ 
32:  $\hat{\mathbf{a}}_b \leftarrow \mathbf{1}\{b \in \mathcal{A}\}, \forall b$ 
33:  $\hat{z}_{i,b} \leftarrow z_{i,b}, \forall i, b$ 
34: return  $\hat{\mathbf{Z}}, \hat{\mathbf{a}}$ 

```

Since concatenation and multiplication by scalar α_ℓ act on feature coordinates and not on UE indices,

$$\|_{\ell \in \mathcal{L}} \alpha_\ell \tilde{\mathbf{H}}^{(\ell)} = \Pi \left(\|_{\ell \in \mathcal{L}} \alpha_\ell \mathbf{H}^{(\ell)} \right). \quad (40)$$

This proves (38). $\square$

Theorem 2 (Non-injectivity of uniform layer aggregation). *Let*

$$C(\mathfrak{M}) = (\mathbf{X}, \bar{\mathbf{S}}), \quad \bar{\mathbf{S}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{S}^{(\ell)}, \quad (41)$$

be the uniform layer aggregation map. If $|\mathcal{L}| \geq 2$, then C is not injective. Hence any predictor $H(C(\mathfrak{M}))$ cannot distinguish all multilayer inputs.

Proof. It is sufficient to construct two distinct inputs with the same image under C . Consider two layers p and q , and let $\mathbf{A} \neq \mathbf{B}$. Define

$$\begin{aligned} \mathfrak{M}_1 : \quad & \mathbf{S}^{(p)} = \mathbf{A}, \quad \mathbf{S}^{(q)} = \mathbf{B}, \\ \mathfrak{M}_2 : \quad & \mathbf{S}^{(p)} = \mathbf{B}, \quad \mathbf{S}^{(q)} = \mathbf{A}, \end{aligned} \quad (42)$$

with all remaining layers and $\mathbf{X}$ identical. Then

$$C(\mathfrak{M}_1) = C(\mathfrak{M}_2), \quad \mathfrak{M}_1 \neq \mathfrak{M}_2. \quad (43)$$

Thus C is non-injective. Consequently, every predictor that factors through C assigns the same output to $\mathfrak{M}_1$ and $\mathfrak{M}_2$. In contrast, the layer-aware encoder contains the ordered terms $\phi(\mathbf{S}^{(p)} \mathbf{X} \mathbf{W}^{(p)})$ and $\phi(\mathbf{S}^{(q)} \mathbf{X} \mathbf{W}^{(q)})$; for $\mathbf{W}^{(p)} \neq \mathbf{W}^{(q)}$, the two constructed inputs need not have identical embeddings. $\square$

Lemma 1 (Attention normalization). *Let α be defined by (29). Then*

$$\alpha \in \Delta^{|\mathcal{L}|-1} = \{\mathbf{u} \in \mathbb{R}_+^{|\mathcal{L}|} : \sum_{\ell \in \mathcal{L}} u_\ell = 1\}. \quad (44)$$

Proof. For all ℓ , $\exp(\eta_\ell) > 0$; therefore $\alpha_\ell \geq 0$. Moreover,

$$\sum_{\ell \in \mathcal{L}} \alpha_\ell = \frac{\sum_{\ell \in \mathcal{L}} \exp(\eta_\ell)}{\sum_{r \in \mathcal{L}} \exp(\eta_r)} = 1. \quad (45)$$

$\square$

The layer-attention concentration reported in the experiments is measured by

$$H_\alpha = - \sum_{\ell \in \mathcal{L}} \alpha_\ell \log \alpha_\ell. \quad (46)$$

Theorem 3 (Finite monotone repair). *Let $\mathcal{A}^{(0)}$ be the active set initialized by (33). Algorithm 1 generates a sequence $\{\mathcal{A}^{(m)}\}_{m \geq 0}$ satisfying*

$$\mathcal{A}^{(0)} \subseteq \mathcal{A}^{(1)} \subseteq \dots \subseteq \mathcal{B}. \quad (47)$$

Moreover, the number of strict expansions is at most $B - |\mathcal{A}^{(0)}|$, and if the algorithm returns at iteration m with $\Phi(\hat{\mathbf{Z}}, \hat{\mathbf{a}}) = 1$, then $(\hat{\mathbf{Z}}, \hat{\mathbf{a}})$ is feasible for (18).

Proof. At any infeasible iteration, the update has the form

$$\mathcal{A}^{(m+1)} = \mathcal{A}^{(m)} \cup \{b_m^+\}, \quad b_m^+ \in \mathcal{B} \setminus \mathcal{A}^{(m)}. \quad (48)$$

Therefore (47) holds and each strict expansion increases $|\mathcal{A}^{(m)}|$ by one. Since $|\mathcal{B}| = B$, at most $B - |\mathcal{A}^{(0)}|$ strict expansions are possible. If the algorithm exits with $\Phi(\hat{\mathbf{Z}}, \hat{\mathbf{a}}) = 1$, then the association, activation, RSRP, rate, and load constraints hold by the definition of Φ . $\square$

Theorem 4 (Switching-aware acceptance). *For a fixed snapshot t and previous active vector $\mathbf{a}(t-1)$, let $J_{\text{sw}}(\mathcal{A}, t)$ be defined by (35). If the stability-aware pruning rule replaces a feasible active set $\mathcal{A}$ by another feasible active set $\mathcal{A}'$ through voluntary pruning, then*

$$J_{\text{sw}}(\mathcal{A}', t) \leq J_{\text{sw}}(\mathcal{A}, t). \quad (49)$$

If the rule activates additional BSs because $\mathcal{A}$ is infeasible, the transition is feasibility driven and is handled by the finite monotone repair property.

Proof. By construction, the stability-aware rule evaluates a candidate feasible set $\mathcal{A}'$ and accepts the replacement only when the switching-adjusted objective in (35) does not increase. Therefore the stated inequality follows directly from the acceptance condition. If $\mathcal{A}$ is infeasible, the replacement is not a voluntary energy pruning step; the algorithm activates additional BSs to recover feasibility, and termination follows from Theorem 3. $\square$

Corollary 1 (Finite-reference bound). *Let $\mathcal{C}$ be the finite candidate set searched by the reference policy and let $J(\mathbf{Z}, \mathbf{a}) = P_{\text{net}}(\mathbf{Z}, \mathbf{a})$. If*

$$(\mathbf{Z}^*, \mathbf{a}^*) \in \arg \min_{(\mathbf{Z}, \mathbf{a}) \in \mathcal{C}, \Phi(\mathbf{Z}, \mathbf{a})=1} J(\mathbf{Z}, \mathbf{a}), \quad (50)$$

then for every feasible $(\mathbf{Z}, \mathbf{a}) \in \mathcal{C}$,

$$J(\mathbf{Z}^*, \mathbf{a}^*) \leq J(\mathbf{Z}, \mathbf{a}). \quad (51)$$

Proof. The inequality is the defining property of the minimum over the feasible subset of $\mathcal{C}$. $\square$

V. EXPERIMENTAL DESIGN

A. Simulation Scenario

The experimental study uses simulated dense downlink cellular snapshots so that the true channel, traffic, association, load, and energy quantities are available for controlled comparison. Each snapshot contains B BSs deployed in a circular service region and N UEs drawn uniformly over the same region. One BS is placed at the center and the remaining BSs are placed on a ring, producing a dense small-cell layout with overlapping coverage. UE traffic demands vary across space and time: each snapshot is assigned a traffic period $t \in \{0, \dots, T-1\}$, and the demand distribution combines lognormal user variation, a periodic traffic factor, and a spatial hotspot.

The downlink channel gain $g_{i,b}$ includes distance-dependent path loss, lognormal shadowing, and small-scale fading. Unless otherwise stated, all BSs transmit with the same power and share the same bandwidth. The all-on network is used only as a reference operating point; learned and heuristic policies are allowed to switch off BSs through the active-BS vector $\mathbf{a}$ defined in Section III. Fig. 3 illustrates the simulated deployment, sleep-cell reassociation effect, and network-side information flow used to evaluate BS ON/OFF decisions. Table I lists the baseline simulation configuration used for the initial comparative study.

B. Reference Decisions

Each training snapshot is labeled by a finite-search reference policy. The reference policy enumerates candidate active-BS subsets $\mathcal{A} \subseteq \mathcal{B}$ and assigns UEs greedily within each subset using channel quality, demand, and projected load. For each candidate, it computes an energy-penalized score

$$Q(\mathbf{Z}, \mathbf{a}) = P_{\text{net}}(\mathbf{Z}, \mathbf{a}) + \lambda_s (1 - \text{SR}(\mathbf{Z}, \mathbf{a})) + \lambda_\rho \left[\max_b \rho_b(\mathbf{Z}, \mathbf{a}) - \rho_{\max} \right]_+, \quad (52)$$

TABLE I
BASELINE SIMULATION AND TRAINING CONFIGURATION

Parameter	Value
Number of BSs, B	7
Number of UEs, N	50
Service-region radius	650 m
Carrier frequency	3.5 GHz
Bandwidth	20 MHz
BS transmit power	20 W
Noise power	−94 dBm
Path-loss exponent	3.2
Shadowing standard deviation	6 dB
Minimum RSRP threshold	−115 dBm
Minimum rate, $R_{\min}$	1 Mbps
Traffic demand range	0.5–6 Mbps
Load limit, $\rho_{\max}$	1.0
BS fixed/dynamic/sleep power	90 W / 55 W / 8 W
BS switching cost, κ	35 W-equivalent per state change
Traffic periods, T	24
Training/test snapshots per seed	60 / 20
Random seeds	11, 13, 17, 19, 23, 29, 31, 37, 41, 43
Hidden dimension	24
Training epochs / patience	80 / 20
Learning rate	0.018

where $[x]_+ = \max(x, 0)$ and SR is the served ratio. The reference decision is the lowest-score feasible candidate in the searched set. This reference is not used as a claimed globally optimal solution of (18); it is used as a strong finite benchmark for supervised training and for measuring the energy gap of learned policies.

C. Compared Policies

The comparative study evaluates four groups of policies.

First, conventional association heuristics are included as non-learning baselines. The RSRP policy assigns each UE to the BS with maximum received power. The SINR policy assigns each UE to the BS with maximum all-on SINR. The load-aware policy sequentially assigns high-demand UEs to BSs that minimize projected load while maintaining radio feasibility when possible. These heuristics are repaired before evaluation using the same QoS/load guard applied to learned policies, so the comparison measures the quality of the association preference rather than failure of a postprocessing step.

Second, explicit sleep-control baselines are included to avoid comparing against association-only heuristics. RSRP-sleep, SINR-sleep, and load-sleep start from the corresponding association rule and greedily deactivate BSs when the resulting repaired association remains feasible and reduces energy. Greedy-sleep is a stronger baseline that uses the same energy-aware assignment rule as the finite-search reference but performs backward one-cell pruning instead of enumerating all active-BS subsets. These baselines are important because they test whether a learned policy is useful beyond hand-designed BS switch-off logic.

Third, single-representation learning baselines are trained on the same reference labels. The flat MLP receives UE features only and does not use graph propagation. The aggregated GCN uses the average support

$$\bar{\mathbf{S}} = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \mathbf{S}^{(\ell)} \quad (53)$$

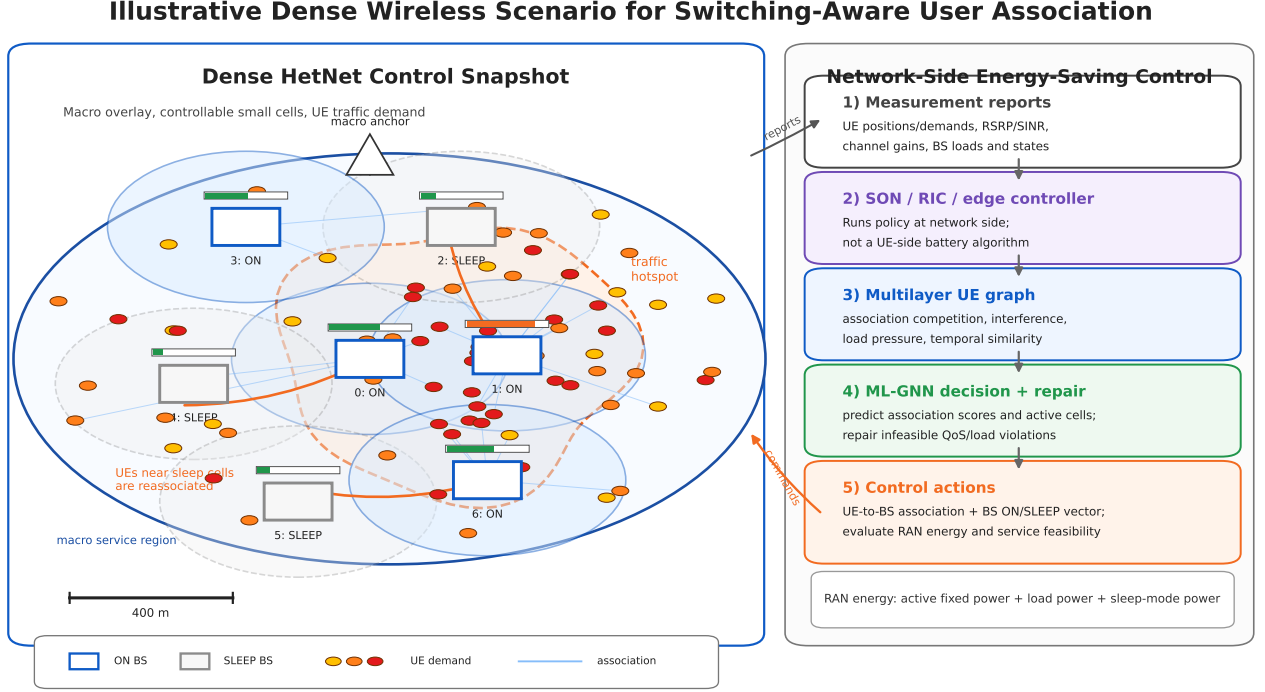


Fig. 3. Illustrative dense wireless control scenario considered in this study. A network-side controller collects radio, traffic, and load information from a dense BS cluster, constructs the multilayer UE graph, and jointly determines UE association and BS active/sleep states. When a BS is switched to sleep mode, its UEs are reassociated to feasible neighboring active BSs subject to QoS and load constraints. The scenario is motivated by HetNet BS switch-off and load-transfer models in [1], [2] and by network-side energy-saving control concepts in O-RAN [5].

and therefore receives a single graph representation. These two baselines separate the value of graph propagation from the value of preserving separate relation layers.

Fourth, the proposed multilayer models are evaluated. The fixed-fusion ML-GCN uses (26)–(27), while the attention ML-GCN uses (29)–(30). For each learned representation, we report the fixed top- K policy, the score-guided pruned policy, and the stability-aware pruned policy. A finite-search reference policy is reported as an upper-bound benchmark within the searched candidate class.

D. Training Protocol

All learning models are trained by minimizing the association loss in (31) over the training snapshots. Node features are standardized using the training set statistics and the same transformation is applied to the test snapshots. The active-BS selector is trained after the association model by using the model probability outputs and the reference active sets. The number of selected active BSs is initialized from the rounded mean active-BS count of the reference decisions in the training set.

For each test snapshot, a learned model first outputs UE-to-BS probabilities and active-BS scores. The decision layer then applies fixed top- K selection, score-guided pruning, or stability-aware pruning. Algorithm 1 finally checks RSRP, rate, and load constraints and activates additional BSs only when the selected set cannot support the snapshot. All reported metrics are computed after this final repaired decision.

E. Evaluation Metrics

The primary metric is network energy. Let P_{all} denote the energy consumed when all BSs are active under the same power model. The energy saving of a policy π is

$$\text{ES}_{\pi} = \frac{P_{\text{all}} - P_{\pi}}{P_{\text{all}}}, \quad (54)$$

where $P_{\pi} = P_{\text{net}}(\mathbf{Z}_{\pi}, \mathbf{a}_{\pi})$. The switching-adjusted energy saving is

$$\text{ES}_{\pi}^{\text{sw}} = \frac{P_{\text{all}} - (P_{\pi} + \kappa S_{\pi})}{P_{\text{all}}}, \quad (55)$$

where S_{π} is the number of BS state changes from the previous snapshot. The static and switching-aware metrics are both reported because a policy can save static energy by changing BS states aggressively, but lose part of that advantage once switching cost is included. The energy gap relative to the reference policy is

$$\text{Gap}_{\pi} = \frac{P_{\pi} - P_{\text{ref}}}{P_{\text{ref}}}, \quad (56)$$

where P_{ref} is the reference energy for the same snapshot.

Constraint satisfaction is measured by served ratio,

$$\text{SR}_{\pi} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left\{ P_{\text{tx}g_{i,b_i}} \geq \Gamma_{\text{rsrp}}, \right. \\ \left. R_{i,b_i} \geq R_{\min}, \rho_{b_i} \leq \rho_{\max} \right\}. \quad (57)$$

Operational compactness and load behavior are reported using active-BS count

$$A_\pi = \sum_{b \in \mathcal{B}} a_{\pi,b}, \quad (58)$$

mean BS state changes

$$\bar{S}_\pi = \frac{1}{T_{\text{test}} - 1} \sum_{t=2}^{T_{\text{test}}} \|\mathbf{a}_\pi(t) - \mathbf{a}_\pi(t-1)\|_0, \quad (59)$$

maximum load

$$\rho_{\max}^\pi = \max_{b \in \mathcal{B}} \rho_b, \quad (60)$$

and mean active load

$$\bar{\rho}_{\text{on}}^\pi = \frac{\sum_{b \in \mathcal{B}} a_{\pi,b} \rho_b}{\sum_{b \in \mathcal{B}} a_{\pi,b}}. \quad (61)$$

Assignment accuracy against reference labels is reported only as a diagnostic, because a policy can have low label agreement while still producing a feasible low-energy association.

F. Layer Ablation and Stress Tests

To evaluate the role of each relation layer, the experiments include single-layer and leave-one-layer-out variants of the ML-GCN. For a layer ℓ , the leave-one-layer-out energy effect is

$$\Delta_\ell^{(-)} = \text{ES}_{\text{full}} - \text{ES}_{-\ell}, \quad (62)$$

and the single-layer contribution is

$$\Delta_\ell^{(+)} = \text{ES}_\ell - \text{ES}_{\text{flat}}. \quad (63)$$

Positive $\Delta_\ell^{(-)}$ indicates that removing layer ℓ reduces energy saving, while positive $\Delta_\ell^{(+)}$ indicates that layer ℓ alone improves over a feature-only model.

Pairwise layer redundancy is measured before training using support-matrix dissimilarity,

$$D_{\ell,r} = \frac{\|\mathbf{S}^{(\ell)} - \mathbf{S}^{(r)}\|_F}{\|\mathbf{S}^{(\ell)}\|_F + \epsilon}, \quad \ell \neq r, \quad (64)$$

where $\epsilon > 0$ avoids division by zero. Attention concentration is measured using (46). In addition to the baseline setting in Table I, stress tests vary traffic intensity, UE density, BS density, and shadowing variance. These sweeps test whether energy savings remain meaningful when load and channel uncertainty change.

VI. RESULTS AND DISCUSSION

A. Static Energy Versus Switching-Aware Control

Table II summarizes the revised ten-seed comparison. The finite-search reference remains the lowest-energy candidate within the searched class. It achieves 62.3% static energy saving and 52.1% switching-adjusted energy saving relative to the all-on network. The important comparison is between greedy sleep control and the learned stable controllers. Greedy sleep is the strongest non-learning policy for static snapshot energy, with 59.7% static saving. However, it changes BS states more frequently, with 3.29 mean state changes per snapshot. Once the switching penalty is included, its saving drops to 48.4%.

The proposed stability-aware ML-GCN achieves 58.4% static saving and 50.4% switching-adjusted saving while maintaining a served ratio of 1.0. It reduces BS state changes to 2.34 per snapshot, approximately 0.95 fewer state changes than greedy sleep. The attention-weighted stable ML-GCN gives a similar result, with 50.4% switching-adjusted saving and 2.35 state changes. Thus, greedy sleep remains preferable if only static snapshot power is counted, whereas the stability-aware learned controllers become preferable when active BS transitions are penalized.

B. Effect of Stability-Aware Pruning

Fig. 5 reports the BS state-change behavior. The stable ML-GCN reduces switching events from 2.89 for ML-GCN-pruned to 2.34, without reducing the served ratio. The paired ten-seed comparison shows that ML-GCN-stable improves switching-adjusted saving over ML-GCN-pruned by 0.0192 on average, wins in 9 out of 10 seeds, and gives a two-sided sign-test p -value of 0.0215. Against the original fixed top- K ML-GCN, the stable variant improves switching-adjusted saving by 0.0799 and wins in all ten seeds.

Relative to greedy sleep, ML-GCN-stable improves switching-adjusted saving by 0.0199 on average and wins in 8 out of 10 seeds. The corresponding sign-test p -value is 0.1094, so the result should be interpreted as a consistent operational advantage in this simulation setting rather than as overwhelming statistical dominance. This distinction is important: the stable learned controller does not beat greedy sleep because it always finds a lower static energy configuration; it beats greedy sleep because it preserves feasibility while reducing unnecessary active-cell transitions.

C. Switching-Cost Sensitivity

Fig. 6 shows the switching-cost sweep. When $\kappa = 0$, greedy sleep has the highest non-reference energy saving because it is optimized for static snapshot power. As κ increases, the ranking changes because policies with higher active-cell churn lose more energy advantage. The mean crossing point between ML-GCN-pruned and greedy sleep is approximately 33.1 W-equivalent per BS state change. At the value used in the main experiments, $\kappa = 35$, ML-GCN-pruned is approximately tied with greedy sleep, while ML-GCN-stable and attention-ML-GCN-stable are clearly above it. This result supports the inclusion of switching cost as a necessary metric for BS switch-off studies.

D. Multilayer Representation and Learned Baselines

The revised results also show a limitation of the current architecture. The stable multilayer models are competitive, but they do not strictly dominate all learned baselines. Flat MLP-stable obtains the highest mean switching-adjusted saving, 50.6%, compared with 50.4% for ML-GCN-stable and attention-ML-GCN-stable. The pairwise difference between ML-GCN-stable and Flat-MLP-stable is small and not statistically significant. Therefore, the evidence should not be framed as a simple claim that multilayer graph learning dominates every alternative.

TABLE II
TEN-SEED PERFORMANCE WITH SWITCHING COST

Policy	Switching saving	Static saving	Switches	Served ratio	Active BSs
Reference search	0.5205 ± 0.0203	0.6228 ± 0.0198	2.965 ± 0.307	1.000	2.590
Flat MLP-stable	0.5056 ± 0.0278	0.5840 ± 0.0247	2.275 ± 0.310	1.000	2.855
Attn-ML-GCN-stable	0.5042 ± 0.0240	0.5852 ± 0.0226	2.350 ± 0.272	1.000	2.840
ML-GCN-stable	0.5039 ± 0.0231	0.5844 ± 0.0198	2.335 ± 0.273	1.000	2.850
Greedy sleep	0.4840 ± 0.0264	0.5973 ± 0.0260	3.285 ± 0.299	1.000	2.790
ML-GCN-pruned	0.4847 ± 0.0250	0.5844 ± 0.0241	2.890 ± 0.273	1.000	2.875
Attn-ML-GCN-pruned	0.4831 ± 0.0276	0.5843 ± 0.0253	2.935 ± 0.348	1.000	2.875
RSRP sleep	0.4537 ± 0.0429	0.5679 ± 0.0383	3.310 ± 0.267	0.9968	3.095
RSRP	0.1793 ± 0.0119	0.1793 ± 0.0119	0.000	0.9954	7.000

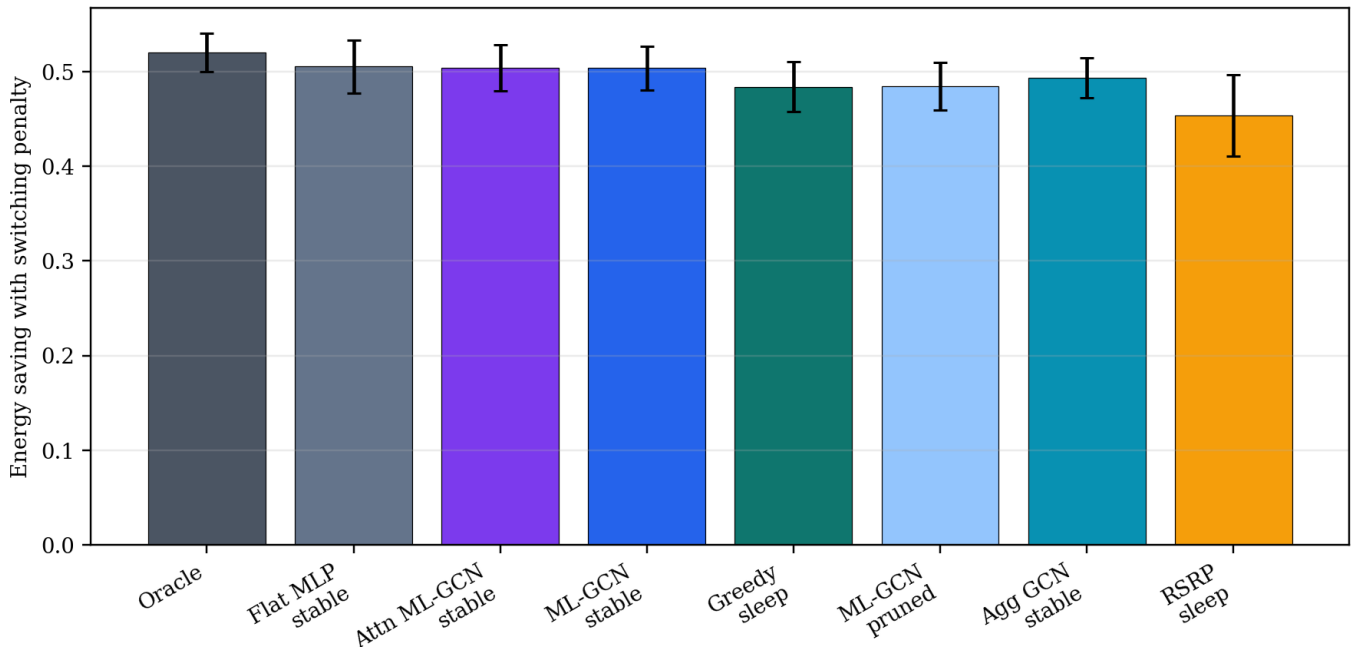


Fig. 4. Switching-adjusted energy saving across the main policies. Stable learned controllers reduce the loss caused by active-BS transitions and outperform greedy sleep under the switching-aware metric.

The defensible interpretation is more specific. The multi-layer representation provides a structured and permutation-equivariant way to model heterogeneous wireless dependencies, while the stability-aware decision layer provides the main operational gain under switching cost. The multilayer stable policies remain close to the best learned controller, preserve full service, and provide relation-aware diagnostics through layer attention and ablation. The current global attention mechanism is conservative; future snapshot-dependent attention may be needed to convert the representational advantage into a larger performance separation.

E. Runtime and Complexity

Table III reports CPU timing and online decision complexity for the revised policies. The benchmark uses $N = 50$ UEs, $B = 7$ BSs, 12 training snapshots, and 5 test snapshots for timing. Let L denote the number of relation layers, d the input feature dimension, h the hidden dimension, and R the

cost of one repair/evaluation pass. Offline training is reported separately because it is amortized before deployment.

The runtime results strengthen the switching-aware interpretation. Greedy sleep is a strong hand-designed controller, but its online latency is higher than ML-GCN-stable and attention-ML-GCN-stable in the timing benchmark. The finite-search reference is substantially slower and scales exponentially with B , so it is best understood as a bounded reference and training signal rather than as an online controller. The stable learned policies therefore provide a middle ground: they are more stable than greedy sleep under switching cost and faster than greedy sleep and finite search at inference time.

F. Discussion

The revised experiments lead to four conclusions. First, association-only heuristics are insufficient for energy-aware control because they do not compact the active-BS set. Second, strong sleep-control baselines must be included; once

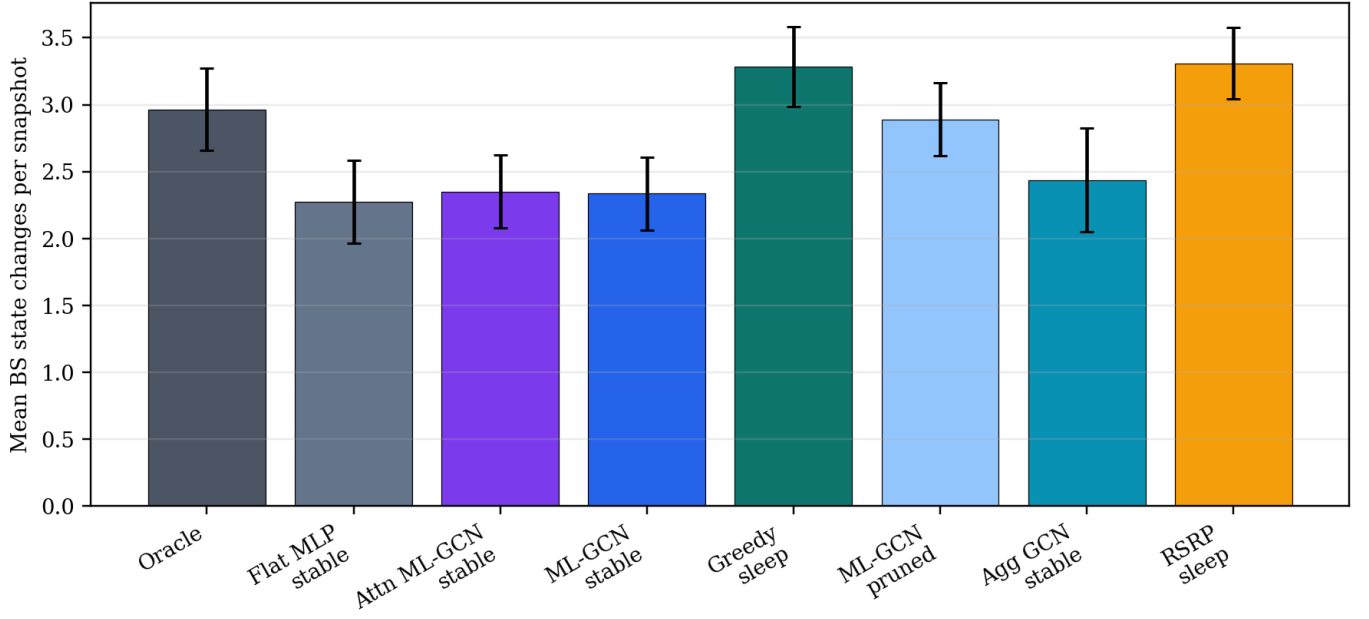


Fig. 5. Mean BS state changes per snapshot. Stability-aware pruning reduces active-cell churn compared with greedy sleep and score-guided pruning.

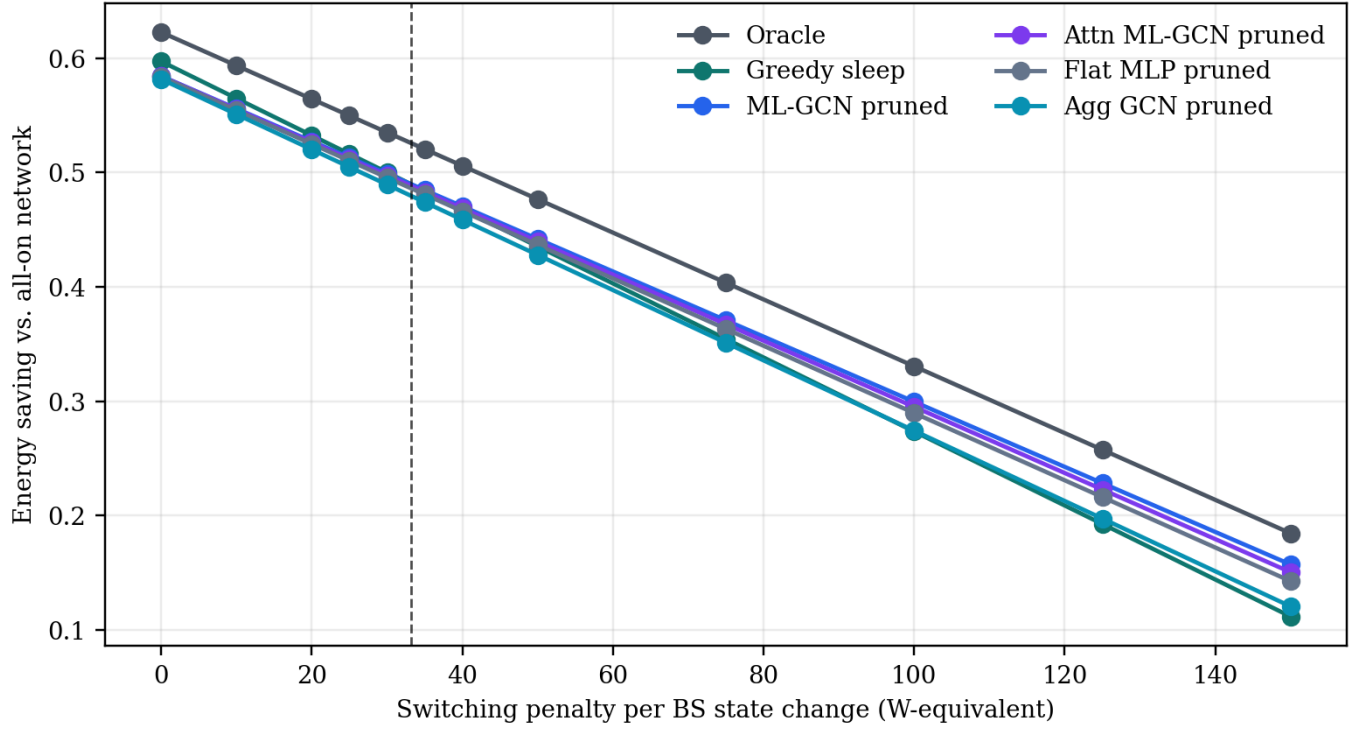


Fig. 6. Sensitivity to the BS switching penalty. Greedy sleep is strongest when switching is free, but learned stable policies become preferable as active-cell transitions become costly.

they are included, greedy sleep is difficult to beat under a static snapshot-energy metric. Third, switching cost changes the operational ranking. Stability-aware learned policies reduce active-cell churn and outperform greedy sleep under switching-adjusted energy while preserving served ratio 1.0. Fourth, the current multilayer GNN should be presented as a

relation-aware component of the learned control framework, not as a model that unconditionally dominates all learned baselines.

This interpretation is stronger and more realistic than a static-energy-only claim. It reflects how BS switch-off would be deployed: a controller must save energy, satisfy QoS

TABLE III
RUNTIME AND ONLINE COMPLEXITY BENCHMARK

Policy	Offline fit (s)	Median latency (ms)	Online decision complexity
RSRP	–	3.48	$\mathcal{O}(NB)$
Load-aware	–	5.13	$\mathcal{O}(NB + N \log N)$
Load-sleep	–	36.16	$\mathcal{O}(B^2N + BN \log N)$
Greedy sleep	–	84.17	$\mathcal{O}(B^2N + BN \log N)$
ML-GCN	0.21	4.98	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B)$
ML-GCN-pruned	0.21	40.06	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B + BR)$
ML-GCN-stable	0.21	51.50	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B + BR)$
Attn-ML-GCN-stable	0.31	41.20	$\mathcal{O}(\sum_{\ell} E_{\ell} d + LNd h + LN h B + BR)$
Reference search	–	228.75	$\mathcal{O}(2^B NB)$

and load constraints, avoid unnecessary BS state changes, and make decisions fast enough for repeated operation. The proposed stability-aware multilayer graph-learning framework addresses these requirements jointly, while also exposing clear directions for improvement in snapshot-dependent attention and constraint-aware training.

VII. CONCLUSION

This paper developed a switching-aware multilayer graph-learning framework for energy-efficient user association and BS switch-off in dense wireless networks. Each wireless snapshot was represented as a UE-level multilayer graph with separate association, interference, load, and temporal relation layers. These layers were processed by fixed-fusion and attention-weighted ML-GCN models, and the learned association preferences were converted into feasible network decisions by active-cell selection, score-guided pruning, stability-aware pruning, and QoS/load repair.

The main finding is that the value of a BS switch-off policy depends on whether active-cell transitions are counted. Under static snapshot energy, greedy sleep control is a strong baseline and achieves the highest non-reference energy saving. Under switching-adjusted energy, however, the stability-aware learned controllers become preferable. The ML-GCN-stable policy achieved 50.4% switching-adjusted energy saving, compared with 48.4% for greedy sleep, while maintaining served ratio 1.0 and reducing BS state changes by approximately 0.95 per snapshot. The stable ML-GCN also improved over the score-guided ML-GCN-pruned policy in 9 out of 10 seeds, with a two-sided sign-test p -value of 0.0215. Runtime measurements showed that the stable learned policies have lower online latency than greedy sleep and substantially lower latency than finite active-cell search.

The results also clarify the scope of the contribution. The multilayer GNN is not claimed to dominate every learned baseline in all metrics. In the present implementation, the stability-aware decision layer is the main driver of switching-adjusted performance, while the multilayer representation provides a relation-aware and permutation-equivariant model for heterogeneous wireless dependencies. This is a more conservative but stronger conclusion: the proposed framework is defensible because it combines graph-based association learning, explicit feasibility repair, active-cell stability, and lower online latency

than greedy sleep, rather than because it wins a static-energy metric alone.

The study has several limitations. First, the evaluation is simulation based and uses controlled channel, traffic, and power-model assumptions. These assumptions support reproducible comparison, but they do not capture all operational details of deployed radio access networks. Second, the finite reference search is tractable for the considered BS scale and is used as a bounded reference rather than as a scalable optimal solver. Third, the current attention mechanism uses global layer weights; it does not yet adapt relation importance on a snapshot-by-snapshot basis. Fourth, the stability-aware pruning rule is a post-learning decision layer, so switching cost is not yet optimized end-to-end inside the GNN training objective.

Future work will extend the framework in four directions. First, the simulation environment should be expanded to include irregular BS deployments, mobility traces, nonuniform traffic hotspots, and more realistic switching delays and signaling costs. Second, adaptive attention and layer-selection mechanisms should be developed so that relation importance can change with traffic load, channel uncertainty, and topology. Third, switching cost and feasibility constraints should be integrated more directly into training through constrained learning, differentiable repair, or safe reinforcement learning. Fourth, transfer across network sizes and operating regimes should be evaluated so that models trained on one deployment can generalize to different densities, bandwidths, and service requirements. These extensions would move the proposed switching-aware multilayer graph framework closer to practical energy-efficient control in future intelligent wireless networks.

DATA AND CODE AVAILABILITY

The source code, simulation configurations, generated result files, MATLAB figure scripts, and manuscript assets used in this study are publicly available at <https://github.com/kazim01011/multilayer-ua-energy>. The repository contains the simulator, multilayer graph construction routines, learning models, baseline policies, reproducibility scripts, and aggregate CSV outputs used to generate the reported tables and figures.

REFERENCES

- [1] H. Ghazzai, M. J. Farooq, A. Alsharoa, E. Yaacoub, A. Kadri, and M.-S. Alouini, "Green networking in cellular HetNets: A unified radio resource management framework with base station ON/OFF switching," *IEEE Transactions on Vehicular Technology*, vol. 66, no. 7, pp. 5879–5893, 2017.
- [2] E. Oh, K. Son, and B. Krishnamachari, "Dynamic base station switching-on/off strategies for green cellular networks," *IEEE Transactions on Wireless Communications*, vol. 12, no. 5, pp. 2126–2136, 2013.
- [3] M. Feng, S. Mao, and T. Jiang, "Base station ON-OFF switching in 5G wireless networks: Approaches and challenges," *IEEE Wireless Communications*, vol. 24, no. 4, pp. 46–54, 2017.
- [4] OECD, "The environmental sustainability of communication networks," OECD, Tech. Rep., 2025. [Online]. Available: https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/02/the-environmental-sustainability-of-communication-networks_66fe3a2a/d1cb2210-en.pdf
- [5] O-RAN Alliance, "Potential energy savings features in O-RAN," White Paper, O-RAN.WP.Potential Energy Savings Features-v01.00, 2025. [Online]. Available: <https://mediastorage.o-ran.org/white-papers/O-RAN.SuFG.Potential%20Energy%20Savings%20Features%20in%20O-RAN%20white%20paper%202025-01.pdf>
- [6] M. Eisen and A. Ribeiro, "Optimal wireless resource allocation with random edge graph neural networks," *IEEE Transactions on Signal Processing*, vol. 68, pp. 2977–2991, 2020.
- [7] Y. Shen, Y. Shi, J. Zhang, and K. B. Letaief, "Graph neural networks for scalable radio resource management: Architecture design and theoretical analysis," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 1, pp. 101–115, 2021.
- [8] M. Kivela, A. Arenas, M. Barthelemy, J. P. Gleeson, Y. Moreno, and M. A. Porter, "Multilayer networks," *Journal of Complex Networks*, vol. 2, no. 3, pp. 203–271, 2014.
- [9] M. De Domenico, A. Sol'e-Ribalta, E. Cozzo, M. Kivela, Y. Moreno, M. A. Porter, S. Gomez, and A. Arenas, "Mathematical formulation of multilayer networks," *Physical Review X*, vol. 3, p. 041022, 2013.
- [10] S. Boccaletti, G. Bianconi, R. Criado, C. I. del Genio, J. Gomez-Gardenes, M. Romance, I. Sendina-Nadal, Z. Wang, and M. Zanin, "The structure and dynamics of multilayer networks," *Physics Reports*, vol. 544, no. 1, pp. 1–122, 2014.
- [11] M. De Domenico, V. Nicosia, A. Arenas, and V. Latora, "Structural reducibility of multilayer networks," *Nature Communications*, vol. 6, p. 6864, 2015.
- [12] M. De Domenico, A. Sol'e-Ribalta, E. Omodei, S. Gomez, and A. Arenas, "Ranking in interconnected multilayer networks reveals versatile nodes," *Nature Communications*, vol. 6, p. 6868, 2015.
- [13] J. Mirzaei, J. Mitra, and G. Poitau, "Graph attention network for optimal user association in wireless networks," 2025.
- [14] M. Grassia, M. De Domenico, and G. Mangioni, "mGNN: Generalizing the graph neural networks to the multilayer case," 2021.
- [15] X.-J. Zhang, X. Ding, H.-F. Zhang, D.-H. Pan, and K. Zhong, "A flexible monitoring framework via dynamic-multilayer graph convolution network," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–11, 2023.
- [16] S. Zandi, K. Korangi, M. Oskarsdottir, C. Mues, and C. Bravo, "Attention-based dynamic multilayer graph neural networks for loan default prediction," *European Journal of Operational Research*, vol. 321, no. 2, pp. 586–599, 2025.
- [17] P. Zhang, Y. Zhang, J. Wang, and B. Yin, "MVMA-GCN: Multi-view multi-layer attention graph convolutional networks," *Engineering Applications of Artificial Intelligence*, vol. 126, p. 106717, 2023.